

FrankScore Psychometric Trait Scoring

End-to-End Example with Five Borrowers

Traits, Behavioural Metadata, and Content (Questions and Scenarios)

1 Introduction

This report illustrates how FrankScore combines three core elements:

- **Psychometric traits:** a set of fifteen behavioural and attitudinal dimensions relevant for creditworthiness.
- **Behavioural metadata:** forty-seven digital interaction features recorded while a borrower completes a psychometric assessment.
- **Content:** questionnaire items and short scenarios that ask about real decisions, plans, and attitudes.

For five fictional borrowers, we assume both content responses and metadata are available, and show how these inputs can be used to derive final trait scores on a normalised scale from zero to one. The goal is not to present final production weights, but to provide a clear, explainable blueprint that can be calibrated with real data.

2 Conceptual Framework

FrankScore uses a three-body model:

1. **Traits** are the latent psychological dimensions we want to estimate, such as Conscientiousness or Honesty.
2. **Content** consists of questions and mini scenarios that ask borrowers about their behaviour, preferences, and attitudes.
3. **Behavioural metadata** consists of how borrowers interact with the assessment itself: how fast they respond, whether they skip questions, how they scroll, and so on.

For each trait:

- Content responses are converted into a *content-based trait score* in the range $[0, 1]$.
- Metadata features are transformed into a *behaviour-based trait score* in the range $[0, 1]$.
- These two components are combined into a final trait value:

$$\text{Trait}_{\text{final}} = \alpha \cdot \text{Trait}_{\text{behaviour}} + (1 - \alpha) \cdot \text{Trait}_{\text{content}},$$

where α is a weight that can later be calibrated with real loan performance data.

3 Psychometric Trait Set

FrankScore currently focuses on the following fifteen traits. Trait names are written in full without abbreviations.

Trait	Short description
Conscientiousness	Carefulness, organisation, reliability, and tendency to complete tasks and follow rules.
Impulsivity	Tendency to act quickly without thinking, preferring short term rewards over better long term outcomes.
Confidence in Financial Management	Belief in one's ability to understand, manage, and repay financial obligations.
Planning Horizon	How far into the future the person normally plans when making business and money decisions.
Self-Control	Ability to resist temptation, delay gratification, and stay focused on commitments.
Locus of Control	Belief that outcomes come mainly from one's own actions rather than from luck or external forces.
Honesty	Tendency to tell the truth and avoid misrepresenting information, even when there is a short term benefit in doing so.
Integrity and Rule Following	Respect for rules, contracts, and agreements, even when enforcement is weak or absent.
Sense of Obligation to Repay	Moral commitment to repay loans and honour agreements, feeling responsible for staying current.
Grit and Perseverance	Ability to continue working and solving problems when business conditions are difficult.
Present Bias and Time Preference	Strength of preference for immediate rewards compared to larger future rewards.
Risk Attitude	Comfort with taking financial risks versus preferring very safe options.
Financial Decision Making Quality	Consistency and accuracy in comparing financial options such as loans and repayment plans.
Spending versus Saving Orientation	Tendency to spend income quickly versus building savings and buffers.
Commitment and Follow Through	Likelihood of completing tasks and fulfilling obligations once they are started.

4 Behavioural Metadata Feature Set

During the digital assessment, the application records a set of behavioural signals. For this design we consider the following forty-seven metadata variables. In practice they are normalised to the $[0, 1]$ range before being used in trait calculations.

Metadata name	Interpretation (informal)
completion_status	Whether the assessment was fully completed.
completion_time_sec	Total time from first question to completion.
sessions_count	Number of sessions used to complete the assessment.
partial_attempts_count	Number of starts that did not reach completion.
return_after_disconnection	Whether the borrower returned after exiting or disconnection.
abandonment_flag	Flag indicating that the assessment was never finished.
off_hours_submission	Whether completion occurred at unusual hours.
device_type	Encoded type of device used (for example, mobile, web).
os_type	Encoded operating system type.
avg_response_time_sec	Average time taken to answer questions.
response_time_std_sec	Variability in response times.
median_response_time_sec	Median response time per question.
min_response_time_sec	Shortest response time observed.
max_response_time_sec	Longest response time observed.
idle_time_total_sec	Total time without interactions.
idle_time_ratio	Fraction of total session time spent idle.
rapid_click_events	Number of extremely fast clicks.
hesitation_time_avg	Average hesitation before the first action on items.
hesitation_time_financial_items	Average hesitation on financial items.
hesitation_time_planning_items	Average hesitation on planning items.
skip_rate	Fraction of items skipped or answered with “I do not know”.
skip_rate_financial_items	Fraction of skipped financial items.
skip_rate_integrity_items	Fraction of skipped integrity and honesty items.
answer_change_rate	Fraction of items where answers were changed.
answer_change_rate_financial_items	Answer change rate on financial items.
answer_change_rate_planning_items	Answer change rate on planning items.
answer_change_rate_integrity_items	Answer change rate on integrity items.
extreme_option_rate	Fraction of times the most extreme option was selected.
inconsistency_score	Rate of logically inconsistent answers across item pairs.
social_desirability_score	Degree to which answers look “too perfect” or highly desirable.
scroll_events_count	Number of scroll events.
scroll_speed_avg	Average scroll speed.
scroll_depth_avg	Average fraction of content scrolled.
scroll_zigzag_score	Degree of up-and-down scrolling.
back_navigation_count	Number of times previous questions were revisited.
page_switch_rate	Rate of switching between pages or sections.

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Metadata name	Interpretation (informal)
correct_financial_decisions	Accuracy on financial decision tasks.
time_spent_financial_scenarios	Time spent on financial scenarios.
time_spent_on_planning_items	Time spent on planning items.
pattern_now_vs_later_score	Fraction of choices favouring “now” over “later” in time preference tasks.
pattern_risky_choices_score	Fraction of risky choices in risk tasks.
retries_on_hard_items	Number of retries on hard items.
hard_items_completed_rate	Fraction of hard items completed.
instruction_read_time	Time spent on instruction screens.
terms_screen_time	Time spent on the terms and consent screen.
terms_scroll_depth	Fraction of the terms text that was scrolled through.
instruction_compliance_flags	Count of explicit instruction violations in the user flow.

5 Content Design: Question and Scenario Types

All questions and scenarios are maintained in an item bank. Each item has an identifier, a main trait tag, an item type (such as Likert, time preference, or scenario), and a scoring rule that maps each possible answer to a score between zero and one.

The main item types used are:

- **Likert statements** with five options from “strongly disagree” to “strongly agree”.
- **Time preference choices**, such as “1,000 birr today versus 1,300 birr in one month”.
- **Risk choice scenarios** comparing safe versus risky options.
- **Honesty and integrity scenarios**, for example, how to react when a lender sends too much money by mistake.
- **Financial decision tasks** that compare loans or repayment plans.
- **Planning horizon questions**, asking how far ahead the borrower usually plans.
- **Spending versus saving questions**, asking whether extra income is first used to buy or to save or repay.

Table 3 gives selected example items used for the borrower scenarios in this report.

Table 3: Example content items and scoring rules

Item id	Item text (simplified)	Main trait	Scoring in $[0, 1]$
Q_CNS_1	“I finish important tasks for my business even when I feel tired.” (1–5)	Conscientiousness	$(r - 1)/4$ for response $r \in \{1, \dots, 5\}$ (positive keying).
Q_SELF_1	“If I decide to save money, I usually succeed in doing it.” (1–5)	Self-Control	$(r - 1)/4$ (positive keying).
Q_GRIT_1	“When my business faces problems, I keep trying different solutions until things improve.” (1–5)	Grit and Perseverance	$(r - 1)/4$ (positive keying).
Q_LOC_1	“My financial situation depends mainly on my own decisions and effort.” (1–5)	Locus of Control	$(r - 1)/4$ (positive keying).
Q_DUTY_1	“If I am late on a payment, I feel very uncomfortable until I catch up.” (1–5)	Sense of Obligation to Repay	$(r - 1)/4$ (positive keying).
Q_CONF_1	“I feel confident that I can manage a loan and make payments on time.” (1–5)	Confidence in Financial Management	$(r - 1)/4$ (positive keying).
Q_INT_1	“I follow rules and agreements even when no one is watching.” (1–5)	Integrity and Rule Following	$(r - 1)/4$ (positive keying).
Q_COMM_1	“Once I agree to do something important, I almost always finish it.” (1–5)	Commitment and Follow Through	$(r - 1)/4$ (positive keying).
Q_PB_1	“1,000 birr today or 1,300 birr in one month?”	Present Bias and Time Preference	Today $\rightarrow 0.0$; one month $\rightarrow 1.0$.
Q_RISK_1	“Invest the loan in a safe project with small profit or a risky project with high profit?”	Risk Attitude	Very safe $\rightarrow 0.3$; safe $\rightarrow 0.5$; risky $\rightarrow 0.8$; very risky $\rightarrow 1.0$.
Q_HON_1	“Lender sends you more money than agreed. What do you do?”	Honesty; Integrity and Rule Following	Return all $\rightarrow 1.0$; wait and see $\rightarrow 0.5$; keep silently $\rightarrow 0.0$.
Q_FDQ_1	Choose the cheaper of two loans with different interest and fees.	Financial Decision Making Quality	Correct choice $\rightarrow 1.0$; wrong choice $\rightarrow 0.0$.
Q_PLANH_1	“How far ahead do you usually plan for your business?”	Planning Horizon	This week $\rightarrow 0.0$; one month $\rightarrow 0.33$; one to three months $\rightarrow 0.66$; more than three months $\rightarrow 1.0$.
Q_SS_1	“When I receive extra income, I first think about buying something I want.” (1–5, reverse)	Spending versus Saving Orientation	Reverse coded: 1 $\rightarrow 1.0$, 5 $\rightarrow 0.0$.

6 Scoring Methodology

6.1 Normalisation

All content item scores are explicitly designed to lie in the interval $[0, 1]$. For Likert items with five response options, positive items use

$$\text{item_score} = \frac{r - 1}{4},$$

where r is the chosen response from one to five. Reverse-keyed items invert the scale.

Behavioural metadata variables are normalised based on observed ranges in the data. In addition, for each trait the direction is aligned so that higher values always correspond to more desirable behaviour for that trait, by transformations such as

$$\text{skip_good} = 1 - \text{skip_rate}, \quad \text{idle_good} = 1 - \text{idle_time_ratio}.$$

6.2 Content-Based Trait Scores

For each trait, a subset of items is tagged in the item bank as measuring that trait. For a given borrower and trait, the content-based score is calculated as the average of the item scores for that trait:

$$\text{Trait}_{\text{content}} = \frac{1}{n} \sum_{j=1}^n \text{item_score}_j,$$

where n is the number of items that map to the trait. In a more advanced implementation, item weights can be non-equal, or item parameters can be estimated using item response theory.

6.3 Behaviour-Based Trait Scores from Metadata

Behaviour-based trait scores are derived from the forty-seven metadata variables. For each metadata feature we define a transformed version that reflects “good” behaviour for particular traits. For example:

$$\begin{aligned} \text{completion_good} &= \text{completion_status}, \\ \text{sessions_good} &= 1 - \text{sessions_count}, \\ \text{idle_good} &= 1 - \text{idle_time_ratio}, \\ \text{skip_good} &= 1 - \text{skip_rate}, \\ \text{hard_good} &= \text{hard_items_completed_rate}, \\ \text{speed_impulsive} &= 1 - \text{avg_response_time_sec}, \\ \text{future_preference} &= 1 - \text{pattern_now_vs_later_score}. \end{aligned}$$

Each trait uses a subset of these transformed features. For example, a simple behaviour-based Conscientiousness score can be defined as

$$\text{Conscientiousness}_{\text{behaviour}} = \frac{\text{completion_good} + \text{sessions_good} + \text{skip_good} + \text{idle_good} + \text{hard_good}}{5}.$$

Honesty can incorporate inconsistency and social desirability measures:

$$\begin{aligned}\text{honesty_inconsistency} &= 1 - \text{inconsistency_score}, \\ \text{honesty_skip_good} &= 1 - \text{skip_rate_integrity_items}, \\ \text{honesty_desirability} &= 1 - \text{social_desirability_score},\end{aligned}$$

and a simple behaviour-based Honesty score becomes

$$\text{Honesty}_{\text{behaviour}} = \frac{\text{honesty_inconsistency} + \text{honesty_skip_good} + \text{honesty_desirability}}{3}.$$

The full implementation extends this approach to use all forty-seven metadata variables across the fifteen traits. Some traits use navigation features (`scroll_speed_avg`, `scroll_zigzag_score`, `back_navigation_count`), others use risk and time preference signals (`pattern_risky_choices_score`, `pattern_now_vs_later_score`), and others draw on reading and compliance behaviour (`instruction_read_time`, `terms_scroll_depth`, `instruction_compliance_flags`).

6.4 Combining Content and Metadata

For each trait, the final score is a weighted combination of the behaviour-based and content-based scores:

$$\text{Trait}_{\text{final}} = \alpha \cdot \text{Trait}_{\text{behaviour}} + (1 - \alpha) \cdot \text{Trait}_{\text{content}},$$

with an initial setting such as $\alpha = 0.6$ for behaviour and 0.4 for content. This weight can later be tuned using real repayment outcomes and reliability analyses.

7 Five Borrower Scenarios

This section introduces five fictional borrowers and illustrates how their content responses and metadata patterns translate into trait scores. The numbers are illustrative but consistent with the scoring logic.

Borrower 1: Sara

Sara is a highly organised, future-oriented borrower who finishes tasks and engages carefully with the assessment. She chooses future rewards in time preference questions, prefers safe business projects, returns extra money immediately in the honesty scenario, and selects the cheaper loan correctly. Her metadata shows full completion in one session, very low skip rates, good focus, and strong reading of instructions and terms.

Borrower 2: Mamo

Mamo shows impulsive behaviour and present bias. In content, he often chooses “now” rather than “later”, gives low conscientiousness answers, is strongly drawn to risky options, and chooses the more expensive loan in the financial decision item. Metadata shows fast responses, many rapid clicks, multiple sessions, higher skip rates, and more inconsistent answers.

Borrower 3: Lulit

Lulit tends to answer modestly about herself (medium scores on several traits), but her behaviour in the assessment is very solid: full completion in one session, almost no skips, low idle time, thoughtful response times, and strong financial accuracy. Metadata lifts her trait scores above what self-report would suggest.

Borrower 4: Abdi

Abdi often gives the “right” answers in content, suggesting very high traits. However, his metadata shows many sessions, larger idle time, more skipped items, higher inconsistency, and very high social desirability. Behaviour-based trait scores down-weight overly optimistic self-report signals for traits like Honesty, Conscientiousness, and Commitment and Follow Through.

Borrower 5: Hanna

Hanna shows very high Grit and Perseverance and a strong Sense of Obligation to Repay, but lower Confidence in Financial Management in self-report. Her behaviour tells a different story: she completes all hard items, retries challenging questions, and demonstrates good financial decision accuracy, although she tends to choose very safe options in risk tasks. Metadata helps correct under-confidence and supports a reliable but cautious borrower profile.

8 Example Metadata Snapshot

Table 4 gives a simplified view of metadata for the five borrowers. All values are scaled between zero and one. For brevity only a subset is shown here; in an implementation, all forty-seven would be included.

Table 4: Illustrative metadata values (subset) for five borrowers

Metadata	Sara	Mamo	Lulit	Abdi	Hanna
completion_status	1.00	0.70	1.00	0.80	1.00
sessions_count	0.10	0.60	0.10	0.70	0.40
idle_time_ratio	0.10	0.40	0.05	0.50	0.15
skip_rate	0.02	0.20	0.00	0.15	0.00
hard_items_completed_ratio	0.95	0.50	0.95	0.60	1.00
avg_response_time_sec	0.55	0.20	0.60	0.85	0.70
rapid_click_events	0.05	0.60	0.05	0.02	0.00
pattern_now_vs_later_score	0.20	0.90	0.20	0.50	0.10
pattern_risky_choices_score	0.40	0.90	0.50	0.60	0.30
correct_financial_decisions	0.90	0.30	0.95	1.00	0.80
time_spent_financial_seconds	0.65	0.30	0.70	0.40	0.60
time_spent_on_planning_seconds	0.30	0.20	0.75	0.30	0.70
skip_rate_integrity_items	0.00	0.10	0.00	0.20	0.00
inconsistency_score	0.05	0.30	0.05	0.30	0.10

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Metadata	Sara	Mamo	Lulit	Abdi	Hanna
social_desirability_score	0.30	0.20	0.25	0.80	0.30
instruction_read_time	0.70	0.10	0.80	0.20	0.60
terms_scroll_depth	0.90	0.30	0.95	0.20	0.80
instruction_compliance_flag	0.00	0.10	0.00	0.20	0.00

In practice, all forty-seven metadata features would be used when computing behaviour-based scores for the fifteen traits. Different traits emphasise different subsets of features (for example, risk tasks for Risk Attitude, time preference tasks for Present Bias and Time Preference, navigation and timing for Impulsivity, reading and compliance features for Integrity and Rule Following).

9 Final Trait Scores for Five Borrowers

Table 5 shows illustrative final trait scores for the five borrowers. These scores reflect a combination of content-based and behaviour-based evidence using the methodology described above.

Table 5: Illustrative final trait scores (scale 0 to 1)

Trait	Sara	Mamo	Lulit	Abdi	Hanna
Conscientiousness	0.97	0.30	0.85	0.55	0.80
Impulsivity	0.20	0.80	0.30	0.40	0.35
Confidence in Financial Management	0.80	0.65	0.70	0.80	0.55
Planning Horizon	0.95	0.20	0.80	0.40	0.75
Self-Control	0.90	0.35	0.75	0.50	0.80
Locus of Control	0.80	0.55	0.75	0.70	0.75
Honesty	0.93	0.55	0.95	0.65	0.95
Integrity and Rule Following	0.92	0.50	0.90	0.60	0.90
Sense of Obligation to Repay	0.95	0.50	0.80	0.70	0.90
Grit and Perseverance	0.90	0.40	0.85	0.55	0.98
Present Bias and Time Preference	0.85	0.15	0.80	0.50	0.90
Risk Attitude	0.55	0.90	0.60	0.65	0.40
Financial Decision Making Quality	0.90	0.25	0.95	0.95	0.80
Spending versus Saving Orientation	0.85	0.20	0.60	0.50	0.85
Commitment and Follow Through	0.95	0.40	0.80	0.60	0.90

For example, Sara shows very high Conscientiousness, high Honesty, high Financial Decision Making Quality, and strong Present Bias and Time Preference in favour of the future. Mamo shows low Conscientiousness, high Impulsivity, strong present bias, and low Financial Decision Making Quality. Lulit and Hanna show strong overall profiles with different strengths

(behavioural reliability versus high grit), while Abdi’s final scores balance high self-report with weaker behavioural evidence.

10 Summary and Next Steps

This report demonstrates how FrankScore can:

- Define a clear set of fifteen psychometric traits relevant to creditworthiness.
- Capture rich behavioural metadata (forty-seven features) from the digital assessment process.
- Design content (questions and scenarios) that maps explicitly to these traits.
- Combine content-based and behaviour-based evidence to obtain explainable trait scores for each borrower.
- Represent these outputs as normalised values from zero to one, ready to feed into credit models.

The next steps for the project are to calibrate the scoring rules using real repayment outcomes, test fairness and stability across subgroups, and integrate the psychometric trait scores into the broader FrankScore credit scoring pipeline.